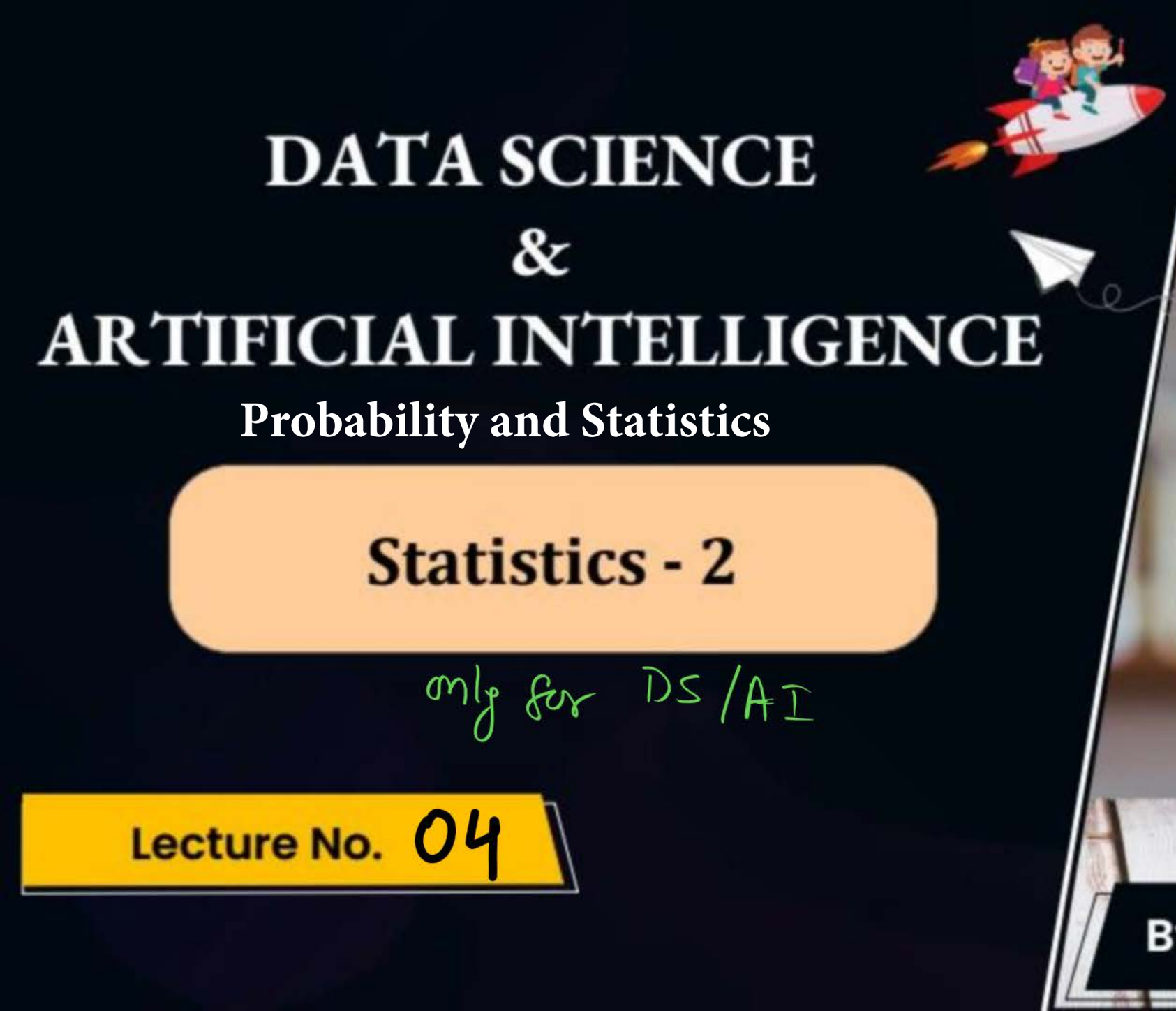
DATA SCIENCE & ARTIFICIAL INTELLIGENCE

Probability and Statistics

Statistics - 2

only for DS/AI

Lecture No. 04



By- Dr. Puneet Sharma Sir

Recap of previous lecture



Topic

BASICS of MULTIPLE INTEGRAL

Topics to be Covered



Topic

BIVARIATE RANDOM VARIABLE

(Continuous Case / joint p.d.f)

Joint p.d.f

Let x & y are C.R.V & their joint p.d.f is $f(x,y)$ then

① $f(x,y) \geq 0$ ② $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$ i.e Total Volume $= 1$

③ $P(a<x<b, c<y<d) = ? = \int_{x=a}^{b}\int_{y=c}^{d} f(x,y)\,dy\,dx$ i.e Volume under $f(x,y)$ b/w $a<x<b$ & $c<y<d$

④ M.D.f dx (V. Strip) is $f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$ & M.D.f dy (H. Strip) is $f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$

⑤ Cond^{n} Density func^{n} dx for given y is $f(x/y) = \frac{f(x,y)}{f(y)} = \frac{\text{joint p.d.f}}{\text{M.D.f } dy}$

& " " " y " " x is $f(y/x) = \frac{f(x,y)}{f(x)} = \frac{\text{joint p.d.f}}{\text{M.D.f } dx}$

(5) $P\{\frac{x=a}{y=b}\} = ? = \frac{P\{x=a \cap y=b\}}{P(y=b)}$

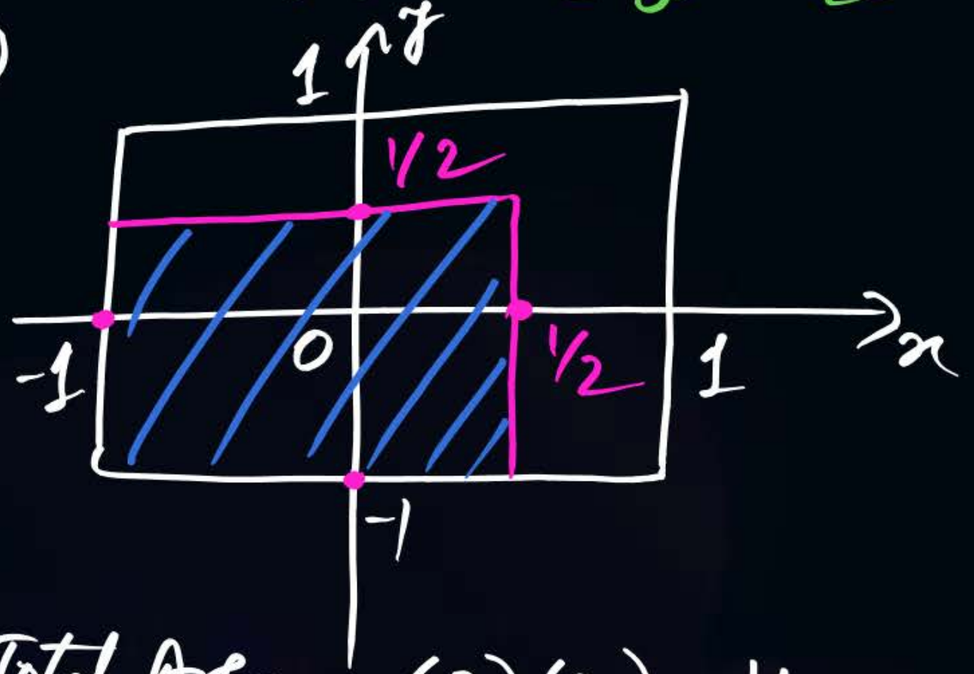
(6) if $[f(x,y) = f(x).f(y) \ \forall \ x \& y]$ $\Longleftrightarrow$ $[x \& y \text{ are Ind.}]$

(7) if $f(x,y)$ is joint p.d.f & $F(x,y)$ is joint C.D.F then

$$F(a,b) = \int_{-\infty}^{a}\int_{-\infty}^{b} f(x,y)\,dy\,dx \quad \& \quad f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

Q: If x & y are two Independent Uniform Random Variables defined in $[-1,1]$ then Find $P\{\max(x,y) < \frac{1}{2}\} = ?$

(M-I)



Total Area $= (2)(2) = 4$

Fav Area $= \left(\frac{3}{2}\right)\left(\frac{3}{2}\right) = \frac{9}{4}$

Hence $P\{\max(x,y) < \frac{1}{2}\}$

$= P\{-1 \le (x,y) < \frac{1}{2}\} = \frac{\text{Shaded area}}{\text{Total area}}$

$= \frac{9/4}{4} = \frac{9}{16}$

(M-II) $\because x \in [-1,1]$ & x is U.R.V, so p.d.f is $f(x) = \begin{cases} \frac{1}{1-(-1)}, & -1 \le x \le 1 \\ 0, & \text{otw} \end{cases}$

Similarly $\because y \in [-1,1]$ & y is also U.R.V, so p.d.f is $f(y) = \begin{cases} \frac{1}{1-(-1)}, & -1 \le y \le 1 \\ 0, & \text{otw} \end{cases}$

$$P\left[\max(x,y) < \tfrac{1}{2}\right] = P\left\{-1 \le (x,y) < \tfrac{1}{2}\right\}$$

$$= P\left\{-1 \le x < \tfrac{1}{2}\right\} \cdot P\left\{-1 \le y < \tfrac{1}{2}\right\} \quad (\because x \& y \text{ are Ind R.V})$$

$$= \int_{-1}^{1/2} f(x)\,dx \cdot \int_{-1}^{1/2} f(y)\,dy$$

$$= \int_{-1}^{1/2} \left(\tfrac{1}{2}\right) dx \int_{-1}^{1/2} \left(\tfrac{1}{2}\right) dy = \tfrac{1}{4}\left(\tfrac{1}{2}+1\right)\left(\tfrac{1}{2}+1\right) = \tfrac{9}{16}$$

M-III

$$f(x)=\begin{cases}\frac{1}{2}, & -1\le x\le 1\\ 0, & \text{otw}\end{cases} \quad \& \quad f(y)=\begin{cases}\frac{1}{2}, & -1\le y\le 1\\ 0, & \text{otw}\end{cases}$$

Since x & y are Ind then their joint p.d.f is $\boxed{f(x,y)=f(x)\cdot f(y)}$

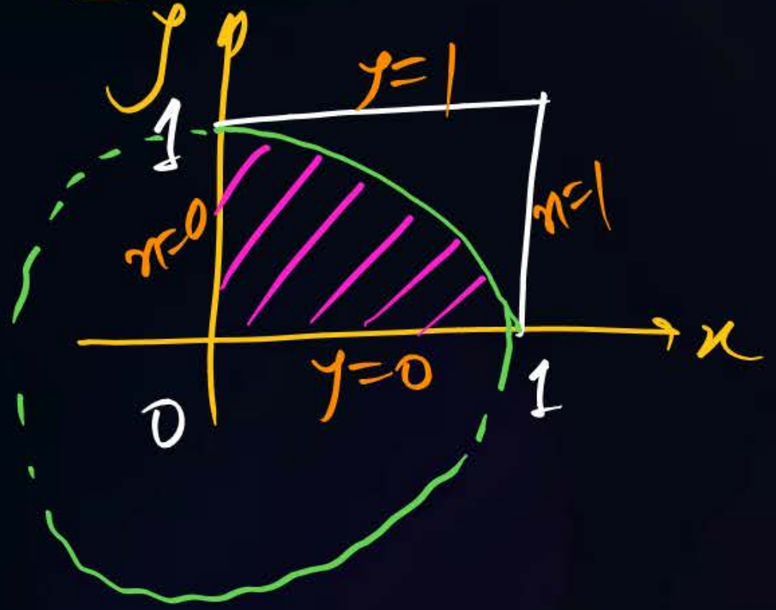
ie $\boxed{f(x,y)=\begin{cases}\frac{1}{4}, & -1\le x\le 1,\ -1\le y\le 1\\ 0, & \text{otherwise}\end{cases}}$

then $P\{\max(x,y)<\frac{1}{2}\}=P\{-1\le(x,y)<\frac{1}{2}\}=\int_{-1}^{1/2}\int_{-1}^{1/2} f(x,y)\,dx\,dy$

$$=\int_{-1}^{1/2}\int_{-1}^{1/2}\left(\frac{1}{4}\right)dx\,dy=\frac{1}{4}\int_{-1}^{1/2}(1)\,dx\int_{-1}^{1/2}(1)\,dy=\frac{1}{4}\times\frac{3}{2}\times\frac{3}{2}=\frac{9}{16}$$

#Q. Two points are randomly selected at random in the interval $[0,1]$. Find the probability that sum of their squares is less than 1.

(M-I) $\because x \in [0,1]$ & $y \in [0,1]$ so Total area of Sq in 1st Quad $= (1)(1) = 1$



fav Area (for $x^2+y^2<1$) $= \frac{1}{4}$ (Area of unit circle)

(ie Interior of unit circle in 1st Quad) $= \frac{1}{4}\{\pi(1)^2\} = \frac{\pi}{4}$

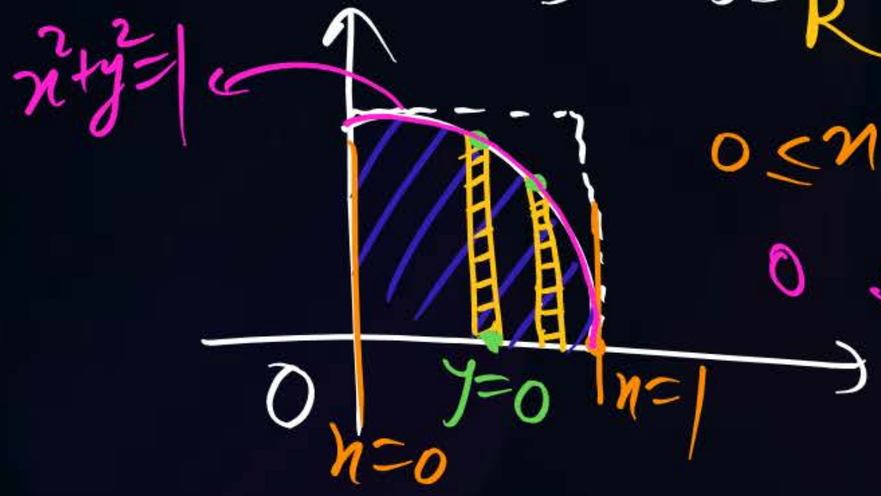
Hence $P\{x^2+y^2<1\} = \frac{\text{fav area}}{\text{Total area}} = \frac{\pi/4}{(1)} = \frac{\pi}{4}$ Ans

(M-II) $\because x \in [0,1]$ & x is U.R.V so p.d.f is $f(x) = \begin{cases} \frac{1}{1-0}, & 0<x<1 \\ 0, & \text{otw} \end{cases}$

$\because y \in [0,1]$ & y is also U.R.V so p.d.f is $f(y) = \begin{cases} \frac{1}{1-0}, & 0<y<1 \\ 0, & \text{otw} \end{cases}$

$\because x$ & y are (Ind) so joint p.d.f is $f(x,y) = f(x).f(y) = \begin{cases} 1, & 0<x<1, 0<y<1 \\ 0, & \text{otw} \end{cases}$

$$P\{x^2+y^2<1\} = \iint_R f(x,y)\,dx\,dy = \int_{x=0}^{1}\int_{y=0}^{\sqrt{1-x^2}} (1)\,dy\,dx = \int_{x=0}^{1}\sqrt{1-x^2}\,dx$$



Put $x = \sin\theta$, $dx = \cos\theta\,d\theta$, $x=0, \theta=0$, $x=1, \theta=\pi/2$

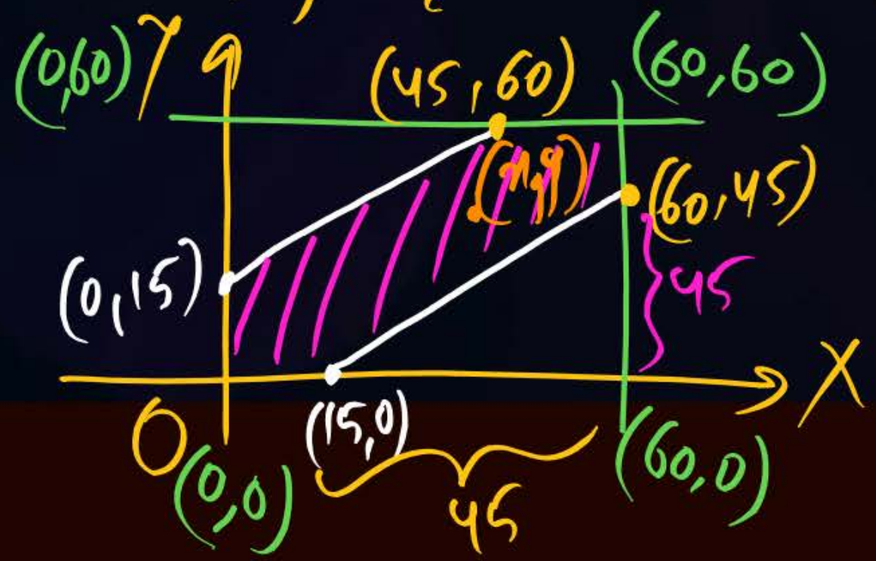
$$= \int_0^{\pi/2} \cos\theta.\cos\theta\,d\theta = \int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right) d\theta = \ldots = \frac{\pi}{4}$$

#Q. A couple wants to meet in park between 1 P.M and 2 P.M, with the understanding that each will wait no longer than 15 minutes for other. What is the probability that they will meet?

HW8 (ii) Find the probability that 1st to arrive has to wait longer than 10 minutes? (25/36)

M-I Let $x = \{$ time of arrival of Boy $\}$ i.e. $x \in (0,60)$ & Total area $= 60 \times 60$

& $y = \{ \,\,\,\, \text{"} \,\,\,\, \text{"} \,\,\,\, \text{"} \,\,\,\, \text{of Girl} \}$ i.e. $y \in (0,60)$



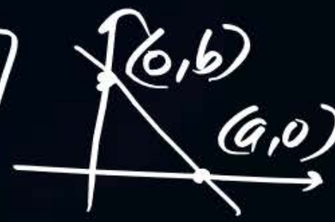
unfav area = 2 area of Lower $\triangle = 2\left[\frac{1}{2} \times 45 \times 45\right]$

& Req Prob $= 1 - P(\text{unfav area})$

$$= 1 - \frac{45 \times 45}{60 \times 60} = 1 - \frac{9}{16} = \frac{7}{16}$$

Note:

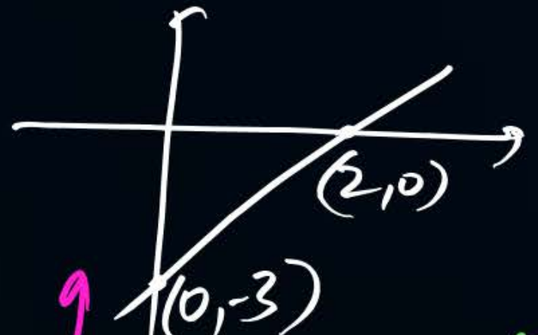
① $\boxed{\frac{x}{a}+\frac{y}{b}=1}$



eg $2y=3x-6$

$3x-2y=6$

$\frac{x}{2}+\frac{y}{-3}=1$



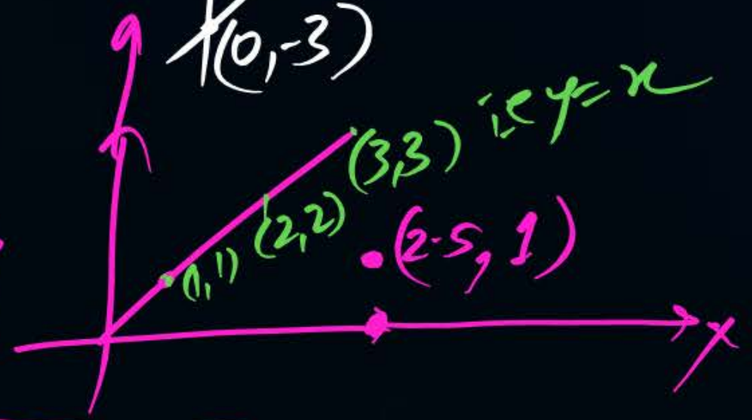
② $\boxed{y=mx}$



③



eg



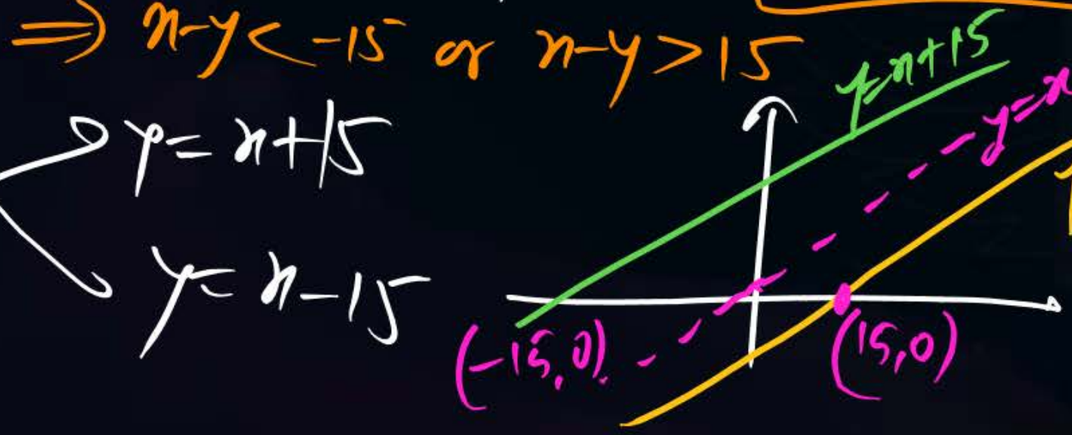
④ $|x-a|<l \Rightarrow -l<(x-a)<l$ or $\boxed{a-l<x<a+l}$

$|x-a|>l \Rightarrow (x-a)<-l$ or $(x-a)>l \Rightarrow \boxed{x<a-l \text{ or } x>a+l}$

eg $|x-y|>15 \Rightarrow x-y<-15$ or $x-y>15$

⑤ $\boxed{y=x}$ $\swarrow y=x+15$

$\searrow y=x-15$



M-II $x = \{$ time of arrival of Male $\} \Rightarrow x \in (0,60)$

$y = \{$ " " " of female $\} \Rightarrow y \in (0,60)$

ATQ, they will meet only when $|x-y| < 15 \Rightarrow -15 < (x-y) < 15$

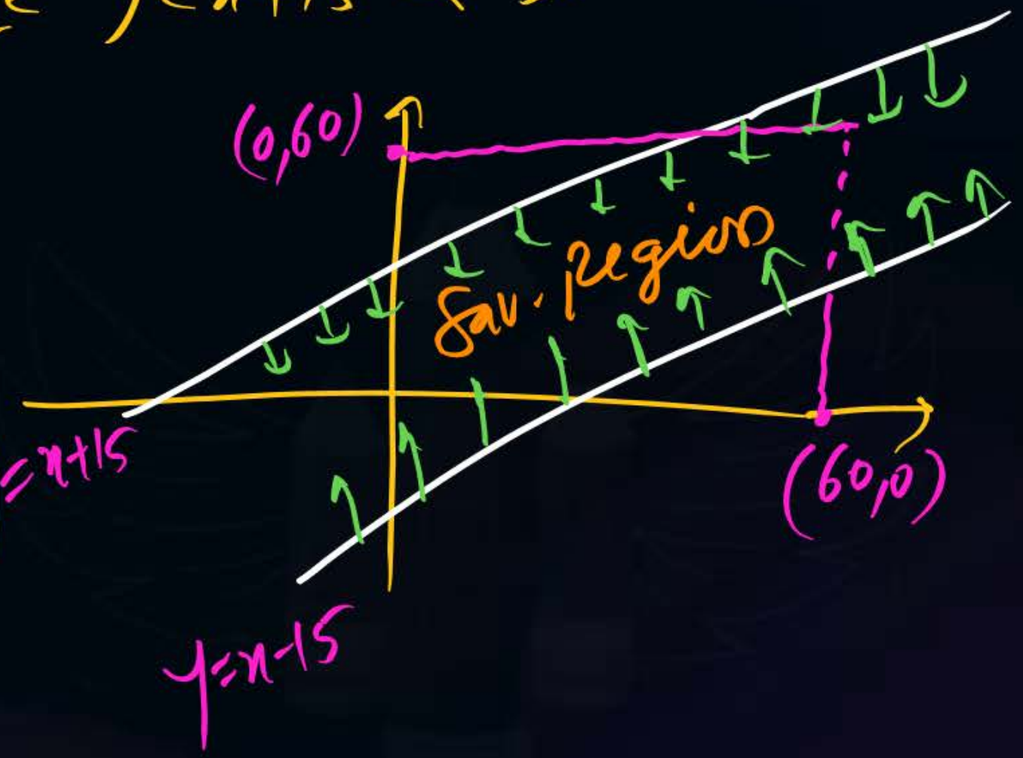
i.e. $y < x+15$ & $y > x-15$

p.d.f of x is $f(x) = \begin{cases} \frac{1}{60}, & 0<x<60 \\ 0, & \text{otw} \end{cases}$

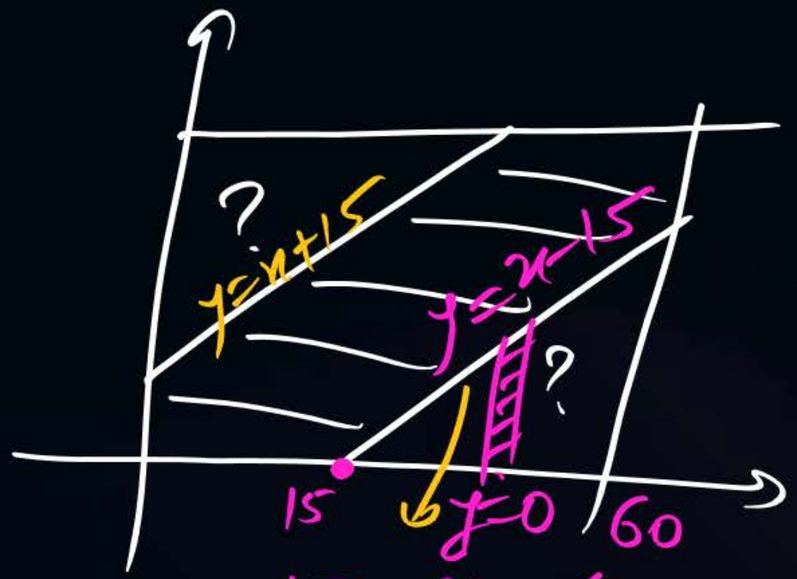
p.d.f of y is $f(y) = \begin{cases} \frac{1}{60}, & 0<y<60 \\ 0, & \text{otw} \end{cases}$

joint p.d.f $f(x,y) = f(x).f(y) = \begin{cases} \frac{1}{3600}, & 0<x<60,\ 0<y<60 \\ 0, & \text{otw} \end{cases}$

$\therefore$ Arrival time of Male & female is Ind.



$P\{\text{they will meet}\} = P\{|x-y| < 15\} = 1 - P\{|x-y| \geq 15\}$



$15 < x < 60$

$0 \leq y \leq x-15$

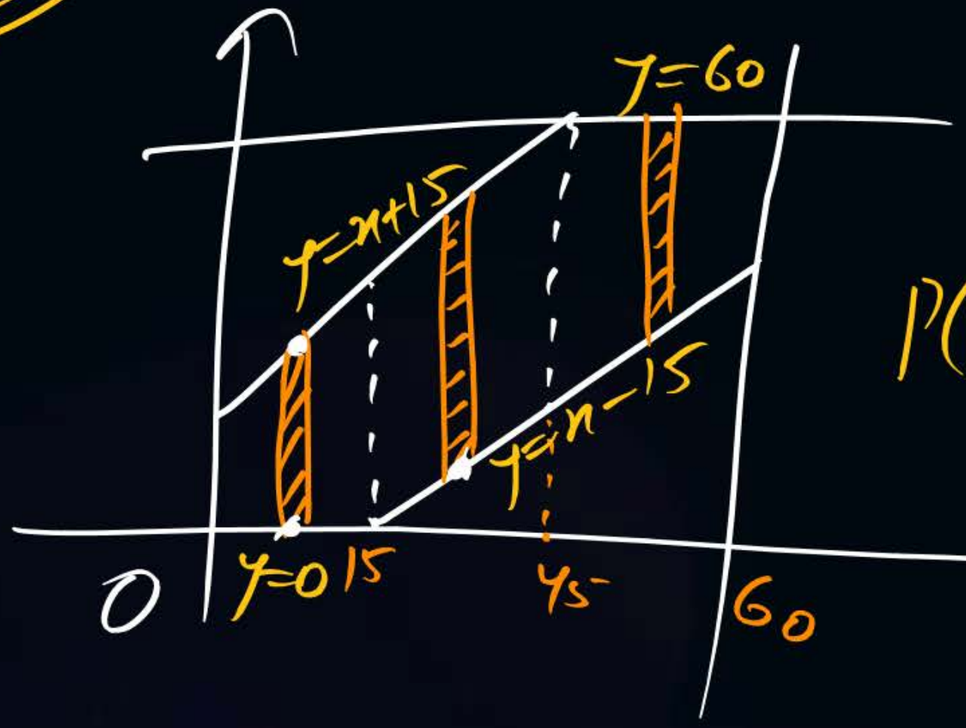
$= 1 - 2P\{(x-y) > 15\}$

$= 1 - 2\int\int f(x,y)\,dx\,dy$

$= 1 - 2\int_{x=15}^{60}\int_{y=0}^{x-15} \left(\frac{1}{3600}\right) dy\,dx$

$= 1 - \frac{2}{3600}\int_{x=15}^{60} (x-15)\,dx = \cdots = \frac{7}{16}$

(n-IV)



$$f(x,y)=\begin{cases}\frac{1}{3600}, & 0<x<60 \\ & 0<y<60 \\ 0, & \text{otw}\end{cases}$$

$$P(|x-y|<15)=\iint f(x,y)\,dx\,dy$$

$$=\int_{x=0}^{15}\int_{y=0}^{x+15}\left(\frac{1}{3600}\right)dy\,dx+\int_{15}^{45}\int_{y=x-15}^{x+15}\left(\frac{1}{3600}\right)dy\,dx+\int_{45}^{60}\int_{y=x-15}^{60}\frac{1}{3600}\,dy\,dx$$

$$=\cdots\cdots=\frac{7}{16}$$

[NAT]

#Q. The joint p.d.f of 2D random variable (x, y) is given as:

$$f(x,y)=\begin{cases}2, & 0<x<1,\ 0<y<x\\ 0, & \text{otherwise}\end{cases}$$

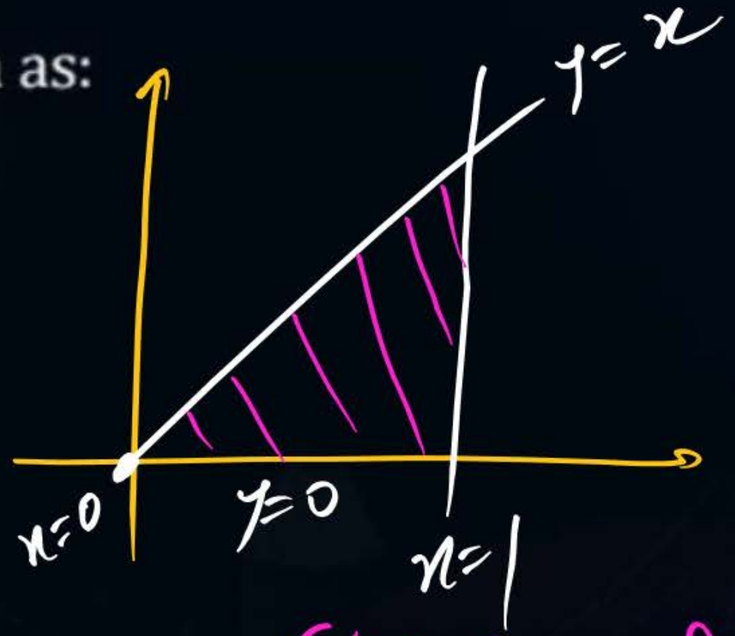
(i) Verify that it is valid p.d.f.

(ii) Find marginal density functions of x and y

(iii) Check for the independence of x and y

(iv) Find the conditional density function of Y for given X = x

(v) Find the conditional density function of X for given Y = y

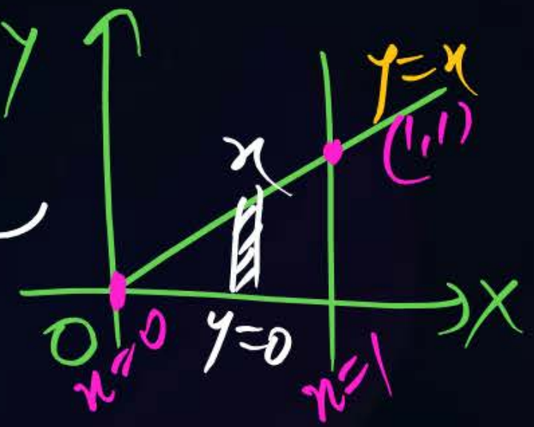


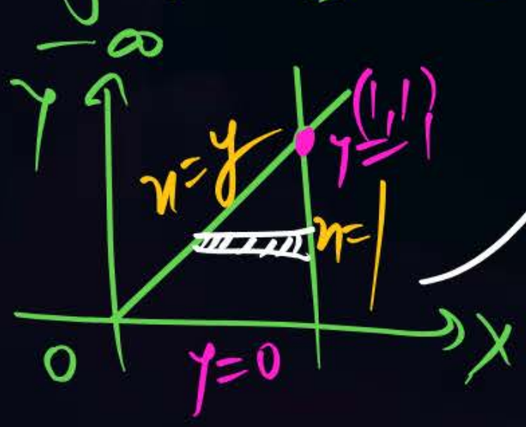
Given Region of Integration

(1) $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = \int_{x=0}^{1}\int_{y=0}^{x} (2)\,dy\,dx = 2\int_{x=0}^{1}\{y\}_{y=0}^{x}.(dx) = 2\int_{x=0}^{1} x\,dx = 1$

(2) M.D.f for x is $f(x) = \int_{y=-\infty}^{\infty} f(x,y)\,dy = \int_{y=0}^{x} (2)\,dy = 2x$; $0<x<1$

M.D.f of y is $f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_{x=y}^{1} (2)\,dx = 2(1-y)$; $0<y<1$





(3) M.D.f of x is $f(x) = 2x,\ 0 \le x \le 1$ & joint p.d.f is

& M.D.f of y is $f(y) = 2(1-y),\ 0 \le y \le 1$

$$f(x,y) = \begin{cases} 2, & 0<x<1, 0<y<x \\ 0, & \text{otw} \end{cases}$$

$\because f(x,y) \neq f(x)\,f(y) \Rightarrow x$ & y are not Independent.

(4) Conditional Density func^n of x for given y is

$$f\left(\frac{X=x}{Y=y}\right) = f\left(\frac{x}{y}\right) = \frac{f(x,y)}{f(y)} = \frac{\text{joint p.d.f}}{\text{M.D.f of } y} = \frac{2}{2(1-y)} = \frac{1}{1-y},\ 0<y<1$$

(5) Conditional Density func^n of y for given x is

$$f\left(\frac{Y=y}{X=x}\right) = f\left(\frac{y}{x}\right) = \frac{f(x,y)}{f(x)} = \frac{\text{joint p.d.f}}{\text{M.D.f of } x} = \frac{2}{2x} = \frac{1}{x},\ 0<x<1$$

[NAT]

#Q. Joint p.d.f. of x and y is given as; $f(x, y) = 4xye^{-(x^2+y^2)}$, $x \geq 0, y \geq 0$

(i) Verify that it is valid p.d.f.

(ii) Find marginal p.d.f. for x and y

(iii) Check the independency of x and y

(iv) Find conditional density of x for given y.



i.e 1st Quadrant is our Region of integration for this Question.

① $f(x,y) = 4xy\, e^{-(x^2+y^2)}$; $x \geq 0, y \geq 0$

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = \int_0^{\infty}\int_0^{\infty} 4xy\, e^{-x^2-y^2}\,dx\,dy$$

$$= \int_{x=0}^{\infty} 2xe^{-x^2}dx \cdot \int_{y=0}^{\infty} 2ye^{-y^2}dy$$

Put $x^2 = \alpha$, $2x\,dx = d\alpha$; Put $y^2 = \beta$, $2y\,dy = d\beta$

$$= \int_0^{\infty} e^{-\alpha}d\alpha \times \int_0^{\infty} e^{-\beta}d\beta$$

$$= \left(\frac{e^{-\alpha}}{-1}\right)_0^{\infty} \times \left(\frac{e^{-\beta}}{-1}\right)_0^{\infty} = (0-1)(0-1) = 1$$ ☺

② M.D.f.(x) is $f(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$

$$= \int_0^{\infty} 4xye^{-x^2-y^2}dy = 2xe^{-x^2}\int_0^{\infty} 2ye^{-y^2}dy$$

$$= 2xe^{-x^2}(1) = \boxed{2xe^{-x^2}}, \quad x \geq 0$$

M.D.f.(y) is $f(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$

$$= \int_0^{\infty} 4xye^{-x^2-y^2}dx = 2ye^{-y^2}\int_0^{\infty} 2xe^{-x^2}dx$$

$$= 2ye^{-y^2}(1) = \boxed{2ye^{-y^2}}, \quad y \geq 0$$

(3) $f(x,y) = 4xye^{-(x^2+y^2)}$ = joint p.d.f

$f(x) = 2xe^{-x^2}, \quad x \geq 0$ M.D.f of x

$f(y) = 2ye^{-y^2}, \quad y \geq 0$ M.D.f of y

$\Rightarrow f(x) \cdot f(y) = f(x,y) \; \forall x \& y$

$\Rightarrow x$ & y are Ind.

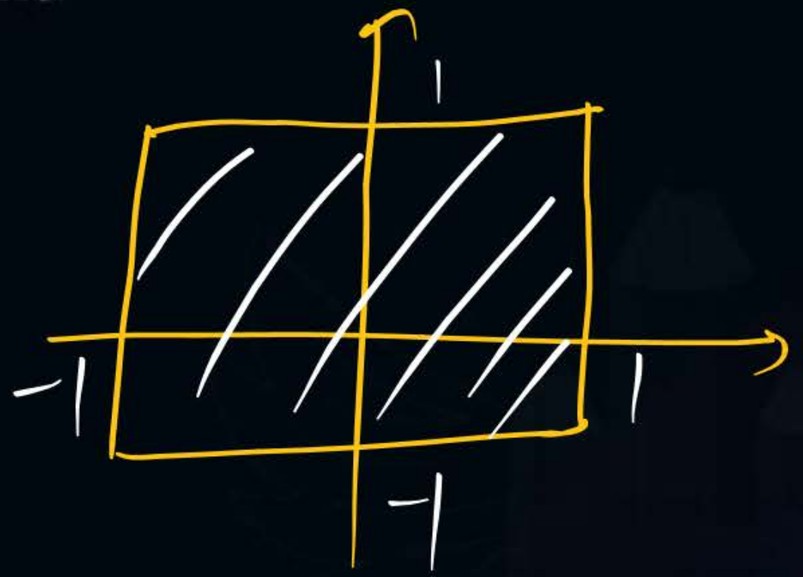
(4) Conditional density of x for given y is

$$f(x/y) = \frac{f(x,y)}{f(y)} = \frac{\text{Joint p.d.f}}{\text{M.D.f of } y} = \frac{4xye^{-(x^2+y^2)}}{2ye^{-y^2}} = 2xe^{-x^2}, \quad x \geq 0$$

[NAT]

#Q. If Joint p.d.f for x and y is given as $f(x, y) = \begin{cases} \frac{1}{4}(1+xy), & |x|<1 \Rightarrow -1<x<1 \\ & |y|<1 \Rightarrow -1<y<1 \\ 0, & \text{otherwise} \end{cases}$

Show that x and y are not independent.



THANK - YOU